Week 3–4 Progress Report
Methodology and System Design

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1 Introduction

This phase focuses on the design and development of an interpretable dataset retrieval system for Open Government Data (OGD) portals. Building upon the limitations identified in previous work, the objective is to construct a retrieval framework that integrates multiple relevance criteria while maintaining transparency in ranking decisions [1].Open Government Data initiatives offer significant benefits but also face adoption barriers such as organizational resistance and data quality issues [2].Open data plays an important role in enhancing transparency and supporting democratic processes when effectively utilized [3].

Traditional keyword-based retrieval approaches are limited by their inability to capture semantic relationships and qualitative aspects of relevance [4]. Although semantic retrieval models improve flexibility, they often lack interpretability due to their complex internal representations [1,5]. In contrast, fuzzy logic provides a structured and interpretable framework for modeling uncertainty and gradual relevance [6].

This phase adopts a Design Science Research (DSR) methodology [7], focusing on the design and implementation of an artifact that addresses the challenges of dataset discoverability.

2 System Architecture

To address the identified limitations in existing retrieval systems, a structured system architecture is required. This architecture supports the integration of multiple metadata features while ensuring transparency and modular design [1].

The proposed system follows a modular architecture that separates query processing, feature extraction, fuzzy inference, and ranking. This design ensures that each component can be independently developed, analyzed, and optimized without affecting the overall system [1].

The query processing module is responsible for transforming raw user input into a structured representation through normalization and enrichment techniques [8]. The feature extraction module computes multiple relevance signals, including keyword similarity, recency, completeness, and category alignment.

These features are then passed to the fuzzy inference system, which integrates them using interpretable fuzzy rules to produce a relevance score [6]. Finally, the ranking module orders datasets based on their computed scores and generates human-readable explanations.

This modular architecture improves system transparency, supports extensibility, and aligns with the principles of Design Science Research by enabling iterative refinement of individual components [7].

2.1 Architecture Diagram

To provide a clear and formal representation of the proposed retrieval framework, the system architecture is presented in Figure 1. The architecture is modeled as a data flow diagram, illustrating the interaction between system components, external data sources, and intermediate data representations.

The system consists of four main modules: query processing, feature extraction, fuzzy inference, and ranking. Additionally, it incorporates an external data source through the CKAN API, which provides structured dataset metadata.

The query processing module transforms raw user input into a normalized query representation. This processed query, together with dataset metadata retrieved via the CKAN API, is passed to the feature extraction module. The feature extraction module computes a feature vector consisting

of multiple relevance signals, including keyword similarity, recency, completeness, and category match.

The resulting feature vector is used as input to the fuzzy inference system, which evaluates fuzzy rules and produces a relevance score. This score is then passed to the ranking module, which orders datasets and generates the final ranked results along with interpretable explanations.

By explicitly modeling data flow, system boundaries, and intermediate representations, the architecture provides a detailed and technically meaningful view of the proposed retrieval system.

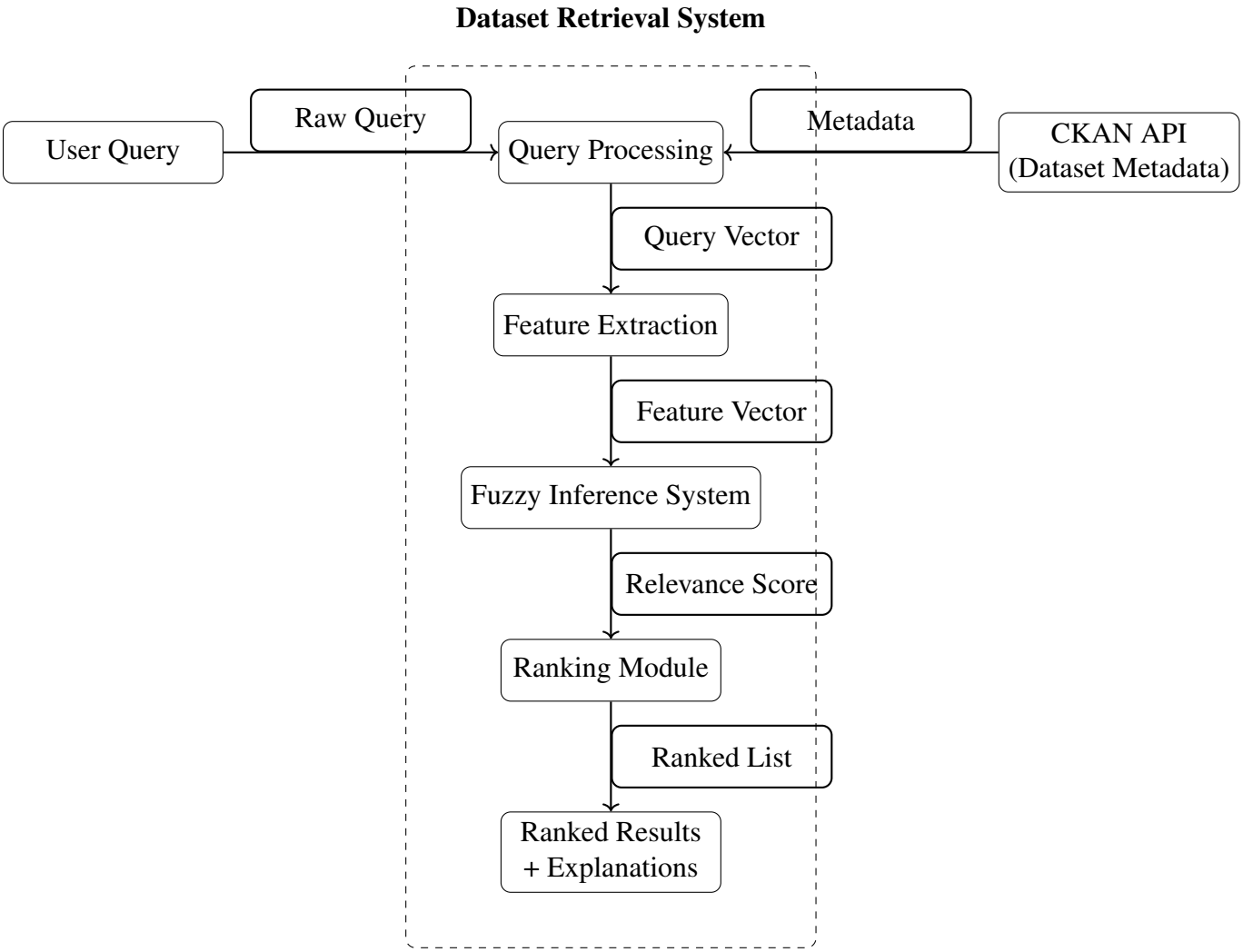

Figure 1: Data flow architecture of the proposed dataset retrieval system showing system boundary, external CKAN API, and intermediate data representations.

Figure 1 illustrates the architecture of the proposed dataset retrieval system, showing the end-to-end processing pipeline from user query input to ranked dataset output. The figure highlights the

interaction between query processing, feature extraction, and the fuzzy inference system, demonstrating how multiple relevance criteria are integrated to produce interpretable ranking results.

2.2 Pipeline Description

The system pipeline transforms a user query into ranked results through multiple stages:

$$q \rightarrow \text{Normalization} \rightarrow \text{Feature Computation} \rightarrow \text{Fuzzy Inference} \rightarrow \text{Ranking} \tag{1}$$

Each stage progressively refines the representation of relevance.

In the normalization stage, the input query is preprocessed using tokenization, lowercasing, and optional synonym expansion to ensure consistency [8]. The feature computation stage then evaluates multiple relevance criteria by comparing the processed query with dataset metadata.

The computed features are passed to the fuzzy inference system, where they are mapped to linguistic variables and evaluated using a set of fuzzy rules [6]. This process enables the integration of multiple criteria in a flexible and interpretable manner.

Finally, the ranking stage converts the fuzzy output into a crisp score through defuzzification and orders datasets accordingly, producing the final ranked list of results.

3 Data Acquisition and Processing

The CKAN API provides structured access to dataset metadata [9], enabling automated retrieval and integration into the system pipeline. This ensures scalability and supports reproducible experimentation across different dataset collections [10]. The CKAN API provides dataset metadata in JSON format through endpoints such as package search. Each response includes fields such as title, description, tags, publication date, and category. These fields are parsed and transformed into structured representations used in the feature engineering stage [9].

Each dataset is represented as:

$$d_i = \{title, description, tags, date, category\} \tag{2}$$

3.1 Preprocessing

Metadata fields are preprocessed using:

- Tokenization
- Lowercasing
- Stopword removal
- Normalization

This preprocessing ensures consistency across datasets [4].

3.2 Query Processing

The query processing module transforms raw user input into a structured representation suitable for feature computation. This step is performed prior to feature extraction to ensure that queries are normalized and enriched in a consistent manner [8].

Query normalization includes the following steps:

- **Tokenization:** The query is split into individual terms, enabling comparison with dataset metadata.
- **Synonym Expansion:** Semantically related terms are incorporated using lexical resources (e.g., WordNet) to improve recall and reduce vocabulary mismatch.
- **Multilingual Normalization:** Queries expressed in different languages are normalized through language detection and translation into a common representation.

The processed query is represented as a normalized term vector:

$$q' = \{t_1, t_2, ..., t_n\} \tag{3}$$

where t_i denotes the processed query terms.

This standardized representation enables consistent feature computation and improves the robustness of the retrieval process, particularly in exploratory search scenarios where queries may be ambiguous or incomplete.

4 Feature Engineering

Feature engineering is critical for capturing multiple dimensions of dataset relevance. In this work, four features are considered: keyword similarity, recency, completeness, and category match.

4.1 Keyword Similarity

Keyword similarity between a query q and a dataset d is computed using the BM25 ranking function, which is a standard and robust method in modern information retrieval.

$$\text{Sim}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d)\,(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \tag{4}$$

where $f(t, d)$ is the frequency of term t in dataset d, $|d|$ is the length of the dataset metadata, and avgdl is the average document length. Parameters k_1 and b control term frequency saturation and length normalization.

Compared to TF-IDF, BM25 provides improved handling of variable-length metadata and term frequency saturation, making it more suitable for dataset retrieval scenarios. BM25 is preferred over TF-IDF due to its ability to account for term frequency saturation and document length normalization [4]. This makes it more robust for datasets with varying metadata lengths, where shorter and longer descriptions should be treated fairly.

4.2 Recency

To address the limitations of linear normalization, recency is modeled using an exponential decay function, which is widely used in information retrieval [4].

$$\text{Recency}(d) = e^{-\lambda \cdot \text{age}(d)} \tag{5}$$

where age(d) represents the age of the dataset in days, and λ is a decay parameter controlling how quickly relevance decreases over time.

This formulation is robust to outliers and preserves discriminative power among recent datasets, unlike max-based normalization which can be distorted by very old datasets.

4.3 Completeness

Completeness is computed as a weighted proportion of available metadata fields:

$$\text{Completeness}(d) = \frac{\sum_{i=1}^{n} w_i \cdot \mathbf{1}_{\{f_i \text{ is present}\}}}{\sum_{i=1}^{n} w_i} \tag{6}$$

where w_i represents the importance weight of metadata field f_i, and $\mathbf{1}$ is an indicator function that equals 1 if the field is present and 0 otherwise.

This weighted formulation accounts for the varying importance of metadata fields. For example, missing critical fields such as title or description has a significantly greater impact on dataset usability than missing auxiliary fields.

4.4 Category Match

Category match is computed using Jaccard similarity between query categories and dataset categories:

$$\text{CategoryMatch}(q, d) = \begin{cases} \frac{|C_q \cap C_d|}{|C_q \cup C_d|}, & \text{if } |C_q \cup C_d| > 0 \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where C_q and C_d represent the sets of categories associated with the query and dataset.

The formulation explicitly handles the edge case where both sets are empty, avoiding division-by-zero errors. Jaccard similarity is chosen due to its effectiveness in comparing categorical sets and its robustness to varying category sizes. Unlike text-based similarity measures, it directly captures overlap between discrete category labels, making it suitable for metadata alignment.

5 Fuzzy Model Design

This section describes the design of the fuzzy inference model used to integrate multiple relevance criteria into a unified and interpretable scoring mechanism.Fuzzy logic has been extensively applied in information retrieval to model uncertainty and partial relevance between queries and documents [11]. The model maps numerical feature values to linguistic variables and applies a set

of fuzzy rules to compute dataset relevance. Special attention is given to ensuring robustness and coverage of the input space to avoid undefined inference outcomes.

5.1 Fuzzy Variables

Each feature is mapped to fuzzy linguistic variables to enable gradual transitions between relevance levels:

- Similarity: Low, Medium, High
- Recency: Old, Moderate, Recent
- Completeness: Poor, Fair, Good
- Category Match: Weak, Moderate, Strong

The output variable is defined as a continuous relevance score in the range $[0, 1]$, representing the overall suitability of a dataset with respect to a given query.

Given four input variables with three linguistic terms each, the complete theoretical rule space consists of $3^4 = 81$ possible combinations. However, defining all combinations manually is impractical and may introduce redundancy.

To address this, a reduced rule base is constructed focusing on the most semantically meaningful combinations. To ensure robustness, a fallback mechanism is introduced to handle cases where no rule is activated.

The selected rules are designed based on the relative importance of features and common retrieval scenarios observed in practice [4]. Instead of exhaustively defining all possible combinations, the rule base focuses on semantically meaningful interactions between features, ensuring interpretability while maintaining sufficient coverage of the input space.

5.2 Fallback Strategy for Incomplete Rule Coverage

Due to the reduced rule base, certain combinations of input features may not activate any fuzzy rule during inference. In such cases, the aggregated firing strength becomes zero, which can lead to undefined behavior in the defuzzification step.

To ensure mathematical robustness, a fallback strategy is introduced. If no rule is activated, a default relevance score is assigned based on a weighted aggregation of the input features:

$$z^* = \sum_{i=1}^{n} w_i \cdot x_i \tag{8}$$

where x_i represents the normalized input feature values and w_i are predefined weights such that $\sum w_i = 1$.

This fallback mechanism guarantees that a valid relevance score is always produced, preventing division-by-zero errors and ensuring stable system behavior.

5.3 Rationale

Fuzzy variables allow the system to model gradual transitions between relevance levels rather than strict boundaries. Unlike Boolean logic, which enforces binary decisions, fuzzy logic captures the inherent uncertainty and subjectivity present in information retrieval tasks [6].

Traditional weighted scoring methods combine features numerically but often lack interpretability, making it difficult to understand how individual criteria influence the final ranking [1]. In contrast, fuzzy logic provides a rule-based framework where decisions can be expressed in human-readable form.

This approach is particularly suitable for dataset retrieval in Open Government Data portals, where relevance depends on multiple partially satisfied criteria [10]. By using fuzzy variables, the system can represent nuanced relationships between features and produce more flexible and interpretable ranking outcomes.

6 Membership Functions

Fuzzy membership functions are used to map numerical feature values to linguistic variables. Triangular membership functions are selected due to their simplicity, interpretability, and computational efficiency [6].

6.1 Triangular Membership Function

The triangular membership function is used to map numerical feature values to linguistic variables. It is defined as:

$$\mu(x; a, b, c) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ \frac{c-x}{c-b}, & b < x < c \\ 0, & x \geq c \end{cases} \tag{9}$$

This function is selected due to its simplicity, interpretability, and computational efficiency.

6.2 Parameter Selection

All input features are normalized to the range $[0, 1]$. Instead of defining membership functions using fixed and uniformly spaced parameters, the boundaries are determined empirically based on the distribution of feature values.

In practice, normalized feature values are often non-uniformly distributed. For example, similarity scores derived from textual representations tend to be right-skewed [4], with most values concentrated in the lower range. Using uniformly spaced membership functions in such cases can lead to poor rule activation, where certain linguistic categories (e.g., "High") are rarely triggered.

To address this issue, the parameters of the triangular membership functions are defined using data-driven quantiles. Let Q_1, Q_2, and Q_3 denote the 25th, 50th (median), and 75th percentiles of a given feature distribution. The membership functions are then defined as:

- Low: $(0, Q_1, Q_2)$
- Medium: (Q_1, Q_2, Q_3)
- High: $(Q_2, Q_3, 1)$

This approach ensures that membership functions are aligned with the actual data distribution, improving the activation of fuzzy rules and enhancing the discriminative power of the model.

The quantiles are estimated from a representative sample of dataset metadata obtained via the CKAN API, ensuring that the membership functions reflect real-world characteristics of the data. This data-driven parameter selection strategy improves robustness and avoids bias introduced by arbitrary parameter choices.

The quantiles used for parameter selection are computed from a representative sample of dataset metadata retrieved via the CKAN API. This ensures that the membership functions reflect real-world data distributions rather than arbitrary assumptions.

7 Fuzzy Rule Base

The fuzzy rule base defines how different features interact to determine dataset relevance. Instead of presenting rules as an unstructured list, the rules are organized into logical groups based on the relative importance of features.

Similarity is treated as the primary relevance signal, followed by recency and metadata quality (completeness and category match). This structured organization improves interpretability and ensures consistent reasoning within the fuzzy inference system.

The rule base is divided into three groups: primary rules, secondary rules, and penalty rules.

7.1 Primary Rules (High Similarity)

These rules capture cases where the dataset strongly matches the query in terms of textual similarity. In such cases, relevance is primarily driven by similarity, with other features acting as supporting signals.

- IF Similarity is High AND Recency is High THEN Relevance is High
- IF Similarity is High AND Completeness is High THEN Relevance is High
- IF Similarity is High AND CategoryMatch is High THEN Relevance is High
- IF Similarity is High AND Recency is Medium THEN Relevance is High

7.2 Secondary Rules (Moderate Similarity)

These rules handle cases where similarity is moderate. In such scenarios, other features such as recency and metadata quality play a more significant role in determining relevance.

- IF Similarity is Medium AND Recency is High THEN Relevance is Medium
- IF Similarity is Medium AND Completeness is High THEN Relevance is Medium
- IF Similarity is Medium AND CategoryMatch is Medium THEN Relevance is Medium
- IF Similarity is Medium AND Recency is Medium THEN Relevance is Medium
- IF Recency is High AND Completeness is Medium THEN Relevance is Medium
- IF Completeness is Medium AND CategoryMatch is Medium THEN Relevance is Medium

7.3 Penalty Rules (Low Quality Conditions)

These rules capture situations where one or more critical features indicate poor dataset quality. In such cases, relevance is reduced regardless of other feature values.

- IF Similarity is Low THEN Relevance is Low
- IF Completeness is Low THEN Relevance is Low
- IF CategoryMatch is Low THEN Relevance is Low
- IF Recency is Low AND Similarity is Medium THEN Relevance is Low

Conflicts between rules are resolved through aggregation using the maximum operator.

8 Fuzzy Inference Mechanism

The system employs the Mamdani fuzzy inference method due to its interpretability and widespread use.

8.1 Inference Steps

1. **Fuzzification:** Convert input features into membership values.

2. **Rule Evaluation:** Apply fuzzy operators (minimum for AND).

3. **Aggregation:** Combine rule outputs using the maximum operator [6].

The firing strength of each rule is computed as:

$$\mu_{\text{rule}} = \min(\mu_1, \mu_2, ..., \mu_n) \tag{10}$$

The Mamdani inference method is selected due to its intuitive rule structure and strong interpretability compared to alternative approaches such as Sugeno inference [6]. While Sugeno models are computationally efficient, they produce outputs that are less transparent, making them less suitable for applications requiring explainability [1].

For example, consider a case where similarity is high and recency is medium. The corresponding fuzzy rule is activated with a firing strength determined by the minimum membership value of the inputs. This firing strength is then used to scale the output fuzzy set, contributing to the final aggregated result.

9 Defuzzification

The centroid method is used to convert the aggregated fuzzy output into a crisp value:

$$z^* = \frac{\int z \cdot \mu(z)\, dz}{\int \mu(z)\, dz} \tag{11}$$

The centroid method computes the center of gravity of the aggregated membership function and is widely used due to its ability to produce smooth and balanced outputs [6]. Unlike alternative methods such as maximum membership, the centroid approach considers the entire output distribution rather than a single point.

This property is particularly important in ranking applications, where small variations in feature values should lead to gradual changes in relevance scores rather than abrupt shifts.

The resulting crisp value represents the overall relevance score of a dataset and is used as the basis for ranking.

10 Ranking Mechanism

Datasets are ranked based on their final defuzzified relevance scores:

$$\text{Rank}(d_i) = \text{argsort}_{d_i}(z_i^*) \tag{12}$$

Datasets are ordered in descending order such that higher relevance scores correspond to higher ranks. This ranking strategy follows standard practices in information retrieval, where documents are sorted based on their estimated relevance to a query [4].

The use of continuous relevance scores enables fine-grained differentiation between datasets, allowing the system to prioritize results that better satisfy multiple criteria.

This approach ensures that the ranking reflects the combined influence of all features integrated through the fuzzy inference system, resulting in a more balanced and interpretable ordering of results.

11 Explainability

The proposed system incorporates explainability as a core design principle by generating human-readable explanations for ranking decisions. Unlike traditional retrieval systems, where explanations often rely on internal model variables, this approach maps inference outcomes back to the underlying dataset metadata to provide meaningful and user-centric justifications [1].

Each dataset's relevance score is influenced by a subset of fuzzy rules activated during inference. Instead of directly exposing linguistic variables such as "high similarity" or "recent," the system translates these signals into explanations grounded in observable dataset attributes.

The explanation generation process consists of two steps. First, the system identifies the most influential features based on their contribution to the final relevance score. Second, these features are mapped back to the original metadata fields from which they were derived.

For example, high similarity is translated into specific matching terms between the query and dataset description, while recency is expressed in terms of the dataset's publication or update date.

An example explanation is:

> "Dataset ranked highly because it strongly matches the query terms 'public transport' in the description and was updated within the last 30 days."

This approach ensures that explanations are grounded in observable dataset attributes rather than abstract linguistic variables, improving user trust and interpretability.

Formally, let x_i denote input features and M_i their corresponding metadata sources. The explanation function $E(d)$ maps the most significant features to metadata-derived statements:

$$E(d) = \{\text{map}(x_i \to M_i) \mid x_i \in \text{TopFeatures}(d)\} \tag{13}$$

This formulation ensures that explanations remain consistent with the model while being grounded in real data.

Explainability is particularly important in Open Government Data systems, where transparency is essential for user trust [10]. By providing explanations that reference concrete dataset attributes, the system enhances interpretability and supports informed decision-making.

12 Prototype Implementation

The proposed dataset retrieval system is implemented as a modular prototype in Python, integrating data acquisition, feature computation, and fuzzy inference into a unified processing pipeline. The implementation is designed to reflect the architecture described in Section 2 and to support extensibility for future evaluation and optimization.

12.1 Implementation Environment

The system is developed using Python due to its extensive support for data processing and scientific computing. The implementation leverages several widely used open-source libraries:

- **pandas**: for handling structured dataset metadata and performing preprocessing operations [12].
- **scikit-fuzzy**: for implementing the fuzzy inference system, including membership functions and rule evaluation [13].
- **NumPy**: for numerical computations and feature representation [14].

The complete implementation of the proposed system is available in an online repository[1].

12.2 Data Integration

Dataset metadata is retrieved from the opendata.swiss portal using the CKAN API. The retrieved data includes attributes such as title, description, tags, publication date, and category. These attributes are stored in a structured format and passed to the feature engineering module for further processing. The use of an API-based data pipeline ensures scalability and supports reproducible experimentation [10].

12.3 Feature Computation Pipeline

The implementation computes multiple relevance features for each dataset:

- Keyword similarity is calculated using BM25 representations of query and dataset text
- Recency is computed based on the age of the dataset
- Completeness is derived from the proportion of available metadata fields
- Category match is evaluated based on alignment between query intent and dataset classification

These features are normalized to a common scale to ensure compatibility with fuzzy membership functions.

[1] GitHub repository: `https://github.com/Deep0901/Master_thesis_project.git`

12.4 Fuzzy Inference Integration

The computed features are passed to the fuzzy inference system implemented using `scikit-fuzzy`. Each feature is mapped to predefined membership functions, and fuzzy rules are applied to compute the relevance score.

The inference process includes fuzzification of input features, evaluation of rule conditions, and aggregation of outputs using Mamdani inference [6]. This integration enables the system to model multi-criteria relevance in an interpretable manner.

12.5 Ranking and Output Generation

The output of the fuzzy inference system is defuzzified using the centroid method to obtain a crisp relevance score for each dataset. Datasets are then ranked in descending order based on these scores.

In addition to ranking, the system generates human-readable explanations by identifying the rules that contributed most significantly to the final score. This supports transparency and enhances user trust in the retrieval process [1].

12.6 System Integration

All components are integrated into a single processing pipeline that takes a user query as input and produces a ranked list of datasets as output. The modular design allows individual components to be modified or extended without affecting the overall system structure.

This prototype implementation demonstrates the feasibility of combining metadata-driven retrieval with fuzzy logic-based ranking and provides a foundation for subsequent evaluation and refinement.

13 Preliminary Validation

To validate the correctness of the system, a set of representative queries is evaluated using a small subset of datasets.

13.1 Results

The system assigns higher relevance scores to datasets that satisfy multiple high-value feature conditions, consistent with multi-criteria retrieval principles [4]. Table 1 presents illustrative examples.

Dataset	Similarity	Recency	Relevance
Dataset A	High	High	High
Dataset B	Medium	Low	Medium
Dataset C	Low	Medium	Low

Table 1: Illustrative validation results

13.2 Interpretability

The system generates explanations based on activated rules, such as:

> ”Dataset ranked highly because it strongly matches the query terms in the description and was recently updated.”

These explanations are consistent with the defined fuzzy rules and reflect interpretable model behavior [1].

13.3 Limitations

The validation is qualitative and limited in scale. A comprehensive quantitative evaluation using standard retrieval metrics such as precision, recall, and MAP is required to fully assess system performance.

14 Limitations

Despite its advantages, the proposed system has several limitations.

First, the design of membership functions is heuristic and may introduce subjectivity [6]. Different parameter choices could lead to variations in ranking behavior.

Second, the rule base requires manual construction, which may limit scalability as the number of features increases. Automated rule learning could address this issue in future work.

Third, the current model assumes equal importance for all features, which may not reflect real-world relevance preferences. Incorporating adaptive weighting mechanisms could improve performance [4].

Finally, the system may not perform optimally for highly complex semantic queries, where deeper contextual understanding is required. Integrating semantic embeddings such as Sentence-BERT could enhance the system's capability in such scenarios [5].

15 Evaluation Plan

The evaluation phase aims to assess the effectiveness and interpretability of the proposed fuzzy logic-based dataset retrieval system. The system will be evaluated by comparing its performance against baseline retrieval approaches using both quantitative and qualitative criteria. The evaluation described in this section is planned for the next phase of the study and is included here to define the experimental methodology.

15.1 Evaluation Objectives

The evaluation is designed to address the following research questions:

- Does the proposed fuzzy retrieval system improve dataset ranking quality compared to traditional keyword-based methods?
- How does the system perform relative to semantic retrieval approaches?
- Does the system provide interpretable and meaningful explanations for ranking decisions?

15.2 Dataset and Query Set

The evaluation will be conducted using datasets obtained from the opendata.swiss portal. A set of representative user queries will be constructed to reflect real-world information needs, including both specific and exploratory queries.

Ground truth relevance labels will be created through manual annotation, where datasets are classified as relevant or non-relevant with respect to each query.

15.3 Baseline Methods

The proposed system will be compared against the following baseline approaches:

- **Keyword-based Retrieval:** TF-IDF or BM25 ranking [4]
- **Semantic Retrieval:** Embedding-based similarity using Sentence-BERT [5]

These baselines represent commonly used retrieval paradigms and provide a benchmark for evaluating performance improvements.

15.4 Evaluation Metrics

The effectiveness of the retrieval system is evaluated using ranking-based metrics designed for graded relevance. Since the proposed fuzzy logic model produces continuous relevance scores in the range $[0, 1]$, while traditional metrics such as precision, recall, and mean average precision (MAP) are widely used, they assume binary relevance judgments. Since the proposed system produces graded relevance scores, nDCG is more suitable for evaluating ranking quality.

To ensure methodological consistency, the evaluation employs Normalized Discounted Cumulative Gain (nDCG), which is specifically designed to assess ranking quality in the presence of graded relevance [15].

15.4.1 Normalized Discounted Cumulative Gain (nDCG)

The Discounted Cumulative Gain (DCG) at rank position k is defined as:

$$\text{DCG@k} = \sum_{i=1}^{k} \frac{2^{rel_i} - 1}{\log_2(i+1)} \tag{14}$$

where rel_i represents the graded relevance score of the dataset at position i.

The ideal DCG (IDCG) is computed by sorting datasets in descending order of their true relevance scores:

$$\text{IDCG@k} = \sum_{i=1}^{k} \frac{2^{rel_i^*} - 1}{\log_2(i+1)} \tag{15}$$

where rel_i^* denotes the relevance score in the ideal ranking.

The normalized metric nDCG is then defined as:

$$\text{nDCG@k} = \frac{\text{DCG@k}}{\text{IDCG@k}} \tag{16}$$

nDCG values range between 0 and 1, where higher values indicate better ranking performance.

This metric is particularly suitable for evaluating the proposed system, as it accounts for both the graded relevance produced by fuzzy inference and the position of datasets in the ranked list. It emphasizes placing highly relevant datasets at top ranks, which aligns with real-world retrieval scenarios.

15.5 Interpretability Evaluation

In addition to retrieval effectiveness, the proposed system is evaluated in terms of interpretability through a user-centered evaluation. Since interpretability and user trust cannot be reliably assessed using offline quantitative metrics alone, a Human-Computer Interaction (HCI) study is designed to evaluate the quality of generated explanations [1, 16].

The evaluation focuses on three key aspects:

- **Consistency:** Whether the explanations accurately reflect the underlying fuzzy inference rules.
- **Understandability:** Whether users can easily comprehend the explanations provided.
- **Usefulness:** Whether the explanations help users understand and trust the ranking decisions.

15.5.1 User Study Design

A user study is conducted with participants familiar with data search tasks. Each participant is presented with a set of queries and corresponding ranked datasets, along with explanations generated by the system.

Participants are asked to evaluate explanations using a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree) based on the following statements:

- “The explanation clearly describes why the dataset was ranked highly.”
- “The explanation is easy to understand.”
- “The explanation increases my confidence in the system’s ranking.”

15.5.2 Evaluation Procedure

The evaluation follows these steps:

1. Select a representative set of queries and datasets.
2. Generate ranked results and corresponding explanations.
3. Present results to participants in a controlled setting.
4. Collect responses using Likert-scale questionnaires.
5. Analyze responses using descriptive statistics (mean and standard deviation).

This user-centered evaluation provides empirical evidence of interpretability by directly measuring user perception and trust, which are essential aspects of explainable systems in public-sector applications.

15.6 Experimental Procedure

The evaluation will follow these steps:

1. Collect datasets and preprocess metadata
2. Define query set and ground truth relevance labels
3. Apply baseline retrieval methods
4. Apply the proposed fuzzy retrieval system
5. Compute evaluation metrics for each method
6. Compare results across different approaches

15.7 Expected Outcomes

The proposed fuzzy logic-based retrieval system is expected to achieve statistically higher nDCG scores compared to baseline methods in both retrieval effectiveness and interpretability. These outcomes will be quantitatively and qualitatively validated using the evaluation methods described in Sections 15.4 and 15.5.

- **Improved Ranking Performance:** The system is expected to achieve higher ranking quality compared to baseline methods such as TF-IDF or BM25. This improvement will be measured using nDCG@k, where the proposed system is expected to yield consistently higher scores across multiple queries.

- **Better Handling of Multi-Criteria Relevance:** By integrating multiple features (similarity, recency, completeness, and category match), the system is expected to provide more balanced relevance estimation. This will be validated by analyzing ranking differences and observing improved positioning of datasets that satisfy multiple criteria simultaneously.

- **Robustness to Diverse Query Types:** The system is expected to perform particularly well for exploratory and ambiguous queries. This will be evaluated by comparing performance across different query categories (specific vs. exploratory) and analyzing variations in nDCG scores.

- **Improved Interpretability and User Trust:** The system is expected to generate explanations that are understandable and useful to users. This will be validated through a user study using Likert-scale responses, where the majority of participants are expected to rate explanations positively in terms of clarity, usefulness, and trust.

- **Consistency Between Model Behavior and Explanations:** The generated explanations are expected to accurately reflect the underlying fuzzy inference process. This will be validated by verifying alignment between activated rules and the corresponding metadata-based explanations.

Overall, these outcomes will provide empirical evidence that combining fuzzy logic with metadata-driven retrieval improves both ranking effectiveness and interpretability in Open Government Data systems.

16 Conclusion

This phase presented the design and development of a fuzzy logic-based dataset retrieval system for OGD portals. The proposed approach integrates multiple metadata-driven relevance criteria, including keyword similarity, recency, completeness, and category alignment, within a unified fuzzy inference framework.

The system architecture was defined as a modular pipeline, enabling the separation of query processing, feature engineering, and ranking logic. A fuzzy model was designed using linguistic variables, membership functions, and a rule-based inference mechanism to support gradual and interpretable relevance estimation. The prototype implementation demonstrated the feasibility of integrating data acquisition, feature computation, and fuzzy inference into a cohesive retrieval system.

Preliminary validation confirmed that the system produces logically consistent rankings and generates human-readable explanations aligned with the defined fuzzy rules. These results indicate that the proposed approach effectively addresses key limitations of traditional and semantic retrieval systems, particularly in terms of interpretability and multi-criteria relevance modeling.

16.1 Next Phase of Work

The next phase of this research will focus on the quantitative and qualitative evaluation of the proposed system. This will involve the construction of a benchmark dataset and query set, along with the development of ground truth relevance annotations.

The evaluation will compare the performance of the fuzzy logic-based retrieval system against baseline methods, including keyword-based and semantic retrieval approaches. Ranking effectiveness will be assessed using graded relevance metrics such as nDCG, which are suitable for evaluating continuous relevance scores produced by the fuzzy inference model [15].

In addition, the interpretability of the system will be evaluated through a user-centered study that analyzes the quality and usefulness of the generated explanations.

The evaluation results will not only provide empirical evidence regarding the effectiveness of the proposed approach but will also inform subsequent refinement of the system. Specifically, observed limitations in ranking performance or explanation quality will be used to adjust membership

functions, update the fuzzy rule base, and refine feature weighting.

This process follows the iterative Design Science Research (DSR) methodology, where evaluation outcomes guide successive improvements of the artifact. The refined model will then be re-evaluated, forming a continuous cycle of design, evaluation, and optimization [7].

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